

# Functional Annotation Report: *sucD* (Q88FB3) — Succinyl-CoA Synthetase $\alpha$ - Subunit of *Pseudomonas putida* KT2440

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## Summary

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The gene **sucD** (ordered locus **PP\_4185**; UniProt **Q88FB3**) of *Pseudomonas putida* KT2440 encodes the  **$\alpha$ -subunit of succinyl-CoA synthetase (SCS)**, also called succinate–CoA ligase [ADP-forming] (EC 6.2.1.5). This enzyme catalyzes the reversible conversion **succinyl-CoA + Pi + ADP  $\rightleftharpoons$  succinate + CoA + ATP** — the single **substrate-level phosphorylation** step of the tricarboxylic acid (TCA/citric acid) cycle. Gene identity is unambiguous: the UniProt record, the KEGG genome annotation, the assigned protein family ("succinate/malate CoA ligase alpha subunit"), and the diagnostic domain architecture (CoA-binding domain IPR003781; CoA\_lig\_alpha IPR005810) all converge on the same well-characterized enzyme, and no conflicting literature for a differently-functioned "sucD" was found for this organism.

Within the two-subunit enzyme, the  **$\alpha$ -subunit (SucD) contributes the two catalytically essential components**: the **CoA-binding site**, located in its N-terminal Rossmann-like domain, and the **catalytic histidine** (His247 in Q88FB3) that is transiently phosphorylated during catalysis to form a phosphohistidine intermediate. The partner  **$\beta$ -subunit (SucC, PP\_4186)** supplies the **dicarboxylate (succinate) binding site** and determines **nucleotide specificity** (ADP/ATP vs. GDP/GTP). The active enzyme assembles as a soluble, cytoplasmic **( $\alpha\beta$ )<sub>2</sub> heterotetramer**. Direct sequence analysis of Q88FB3 confirms the conserved catalytic phosphohistidine motif and the N-terminal glycine-rich CoA/Rossmann motif, and a very-high-confidence AlphaFold model (mean pLDDT 96.5) supports the canonical two-domain SCS- $\alpha$  fold in the actual *P. putida* protein.

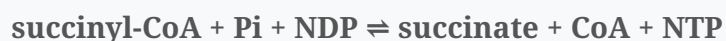
Functionally, SucD operates in the **cytoplasm** as part of the TCA cycle, contributing to central carbon metabolism, energy generation, and — through the succinate/succinyl-CoA node — connections to propanoate metabolism and C5-branched dibasic acid metabolism. In pseudomonads the enzyme has notably **broad NTP-synthesizing capacity** and physically couples with **nucleoside diphosphate kinase (Ndk)**, positioning it as a contributor to cellular nucleotide-triphosphate pools beyond just ATP. This report is built on a four-tier evidence base:

(1) the intrinsic sequence/structure of Q88FB3, (2) KEGG genome context in *P. putida* KT2440, (3) direct biochemical work in the close relative *P. aeruginosa*, and (4) mechanistic crystallographic studies of orthologous SCS enzymes from *E. coli*, pig heart, and *Thermus aquaticus*.

## Key Findings

### Finding 1 — *sucD* encodes the $\alpha$ -subunit catalyzing the TCA cycle's only substrate-level phosphorylation

The gene **sucD (Q88FB3 / PP\_4185)** encodes the  $\alpha$ -subunit of succinyl-CoA synthetase, assigned EC 6.2.1.5 (Succinate–CoA ligase [ADP-forming]). SCS catalyzes the reversible reaction:



This is the **only reaction in the citric acid cycle that provides substrate-level phosphorylation** — that is, the direct synthesis of a nucleoside triphosphate coupled to a chemical bond transformation rather than to the electron-transport chain. The **[ADP-forming]** designation (EC 6.2.1.5) means the *P. putida* enzyme generates **ATP** (from ADP), in contrast to the GTP-forming isoform (EC 6.2.1.4) that uses GDP.

This defining catalytic role is directly stated in the literature: succinyl-CoA synthetase "catalyzes the only step in the citric acid cycle that provides substrate-level phosphorylation" (PMID: 27487822), and the enzyme "catalyses the reversible conversion of succinyl-CoA and ADP or GDP to succinate and ATP or GTP" (PMID: 18392745). The reversibility is physiologically important: in the oxidative direction the enzyme conserves the energy of the succinyl-CoA thioester bond as a phosphoanhydride bond in NTP, while in the reverse direction it can activate succinate to succinyl-CoA.

## Finding 2 — The $\alpha$ -subunit carries the CoA-binding site and the transiently phosphorylated catalytic histidine

Crystallographic dissection of SCS orthologs establishes that the  **$\alpha$ -subunit (SucD) contributes the two components most central to catalysis:**

1. **The CoA-binding site.** In *E. coli* SCS, "CoA is bound in an extended conformation to the nucleotide-binding motif in the N-terminal domain of the alpha-subunit" (PMID: 9917402). The GTP-specific pig-heart structure confirms this: "CoA bound to the amino-terminal domain of the  $\alpha$ -subunit, with the free thiol extending from the adenine portion into the site where the catalytic histidine residue resides" (PMID: 26249701).
2. **The catalytic histidine (phosphohistidine intermediate).** During the reaction a histidine of the  $\alpha$ -subunit is transiently phosphorylated, forming a covalent phospho-enzyme intermediate. This mobile histidine must **shuttle the phosphoryl group** between the CoA/succinate site and the nucleotide site — indeed "the catalytic histidine residue would have to move about 35 Å to react with the nucleotide" (PMID: 9917402). The phosphoryl group is stabilized at the amino termini of two "power" helices, one from the C-terminal domain of the  $\alpha$ -subunit and one from the  $\beta$ -subunit.

Critically, **nucleotide specificity does NOT reside in the  $\alpha$ -subunit**. The ATP-specific and GTP-specific isoforms of pigeon SCS "incorporate the same alpha-subunit," so "the determinants of nucleotide specificity must reside within the beta-subunit" (PMID: 9765290). This means SucD provides the invariant catalytic machinery (CoA binding + phosphohistidine), while its partner SucC ( $\beta$ ) tunes which nucleotide is used.

## Finding 3 — Strict substrate specificity for succinate/succinyl-CoA, with the dicarboxylate site on the $\beta$ -subunit

The enzyme is highly selective for its C4-dicarboxylate substrate. The SCS-succinate-Mg·CoA complex shows that "succinate binds in the carboxy-terminal domain of the  $\beta$ -subunit," and this "succinate-binding site is near both the active-site histidine residue that is phosphorylated in the reaction and the free thiol of CoA" (PMID: 27487822). The architecture positions the  $\alpha$ -subunit's CoA thiol and catalytic histidine immediately adjacent to the  $\beta$ -subunit's dicarboxylate pocket, allowing the thioester-forming/phosphoryl-transfer chemistry to proceed.

The specificity is strict: the structure "shows why succinyl-CoA synthetase is specific for succinate and does not react appreciably with citrate nor with the other C4-dicarboxylic acids of the citric acid cycle, fumarate and oxaloacetate, but shows some activity with L-malate"

([PMID: 27487822](#)). Thus, although SucD supplies the universal CoA moiety, the enzyme as a whole discriminates strongly in favor of succinate over other TCA-cycle dicarboxylic acids — a property conferred largely by the  $\beta$ -subunit's binding pocket and its domain-closure rearrangement upon substrate binding.

#### Finding 4 — Encoded in the *sucCD* operon; functions as a cytoplasmic ( $\alpha\beta$ )<sub>2</sub> tetramer coupled to Ndk with broad NTP-synthesizing capacity

In the close relative *Pseudomonas aeruginosa*, the ***sucCD* operon encodes the  $\alpha$  (*sucD*) and  $\beta$  (*sucC*) subunits** of SCS ([PMID: 10671455](#)). The *P. aeruginosa* enzyme displays **remarkably broad nucleotide usage**: "In the presence of inorganic orthophosphate (Pi), succinyl-CoA, and either GDP, ADP, UDP or CDP, it synthesizes GTP, ATP, UTP, or CTP" ([PMID: 10671455](#)). Furthermore, the *Pseudomonas* SCS physically and functionally couples with **nucleoside diphosphate kinase (Ndk)**, modulating its NTP-synthesis specificity — positioning SCS as a broader contributor to the cellular NTP pool than the canonical ATP-only description would suggest.

Structurally, bacterial SCS assembles as a **heterotetramer of two  $\alpha$ - and two  $\beta$ -subunits** — for *T. aquaticus* SCS, "the enzyme is a heterotetramer of two  $\alpha$ -subunits and two  $\beta$ -subunits" ([PMID: 22751660](#)). As a citric-acid-cycle enzyme, SCS operates in the **bacterial cytoplasm**; in eukaryotes the orthologous enzyme is a **mitochondrial matrix enzyme** ([PMID: 18392745](#)), the compartment that houses the TCA cycle. For *P. putida*, the cytoplasm is therefore the site of SucD's function.

#### Finding 5 — Q88FB3's own sequence confirms the catalytic phosphohistidine (His247) and N-terminal CoA-binding domain

Analysis of the actual *P. putida* protein sequence — rather than reliance on orthologs alone — confirms the SCS- $\alpha$  mechanism applies here. The UniProt Q88FB3 record (294 aa; KEGG ppu:PP\_4185; EMBL AAN69766.1) annotates:

- An **active site at His247**, described as a "*Tele-phosphohistidine intermediate*" — the catalytic residue that is transiently phosphorylated. Direct sequence inspection confirms residue 247 is histidine, embedded in the conserved motif **...PAGKRMG-H247-AGAIISGG...** (the diagnostic G-H-A-G phosphohistidine signature). His247 is the positional equivalent of the experimentally characterized catalytic His246 $\alpha$  of *E. coli* SCS.

- A **CoA-binding domain spanning residues 4–100**, with the N-terminus containing the glycine-rich CoA/Rossmann motif **G16-F-T-G-S-Q-G22** (coincident with annotated binding site 17–20), plus additional ligand-binding residues at 43, 96–98, and 159.

The protein matches the diagnostic PROSITE signatures **PS01216 (SUCCINYL\_COA\_LIG\_1)** and **PS00399 (SUCCINYL\_COA\_LIG\_2)** and Pfam families **PF02629 (CoA\_binding)** + **PF00549 (Ligase\_CoA)**. The literature confirms this catalytic residue is on the  $\alpha$ -subunit: "During the reaction, a histidine residue of the alpha-subunit is transiently phosphorylated" ([PMID: 11781092](#)), and the N-terminal CoA-binding motif matches the "nucleotide-binding motif in the N-terminal domain of the alpha-subunit" ([PMID: 9917402](#)).

## Finding 6 — Genomic context: a canonical TCA gene cluster in *P. putida* KT2440

KEGG genome annotation places **PP\_4185 (sucD)** — ortholog **K01902** ("succinyl-CoA synthetase alpha subunit [EC:6.2.1.5]") — **immediately adjacent to PP\_4186 (sucC)**, ortholog K01903 ("succinyl-CoA synthetase beta subunit"), followed by **PP\_4187 (lpdG / dihydrolipoyl dehydrogenase, K00382)**. This is a canonical TCA-cluster arrangement. sucD is assigned to **KEGG module M00011** (Citrate cycle, second carbon oxidation: 2-oxoglutarate = oxaloacetate) and to pathways:

| KEGG pathway | Description                         |
|--------------|-------------------------------------|
| ppu00020     | Citrate cycle (TCA cycle)           |
| ppu00640     | Propanoate metabolism               |
| ppu00660     | C5-branched dibasic acid metabolism |
| ppu01200     | Carbon metabolism                   |

Notably, the  $\beta$ -subunit PP\_4186 additionally carries the annotation "glutaryl-CoA synthetase, beta subunit," hinting at a broader dicarboxyl-CoA metabolic context in this organism. This directly confirms — in *P. putida* KT2440 itself — the operonic pairing of sucD/sucC and the assignment of SucD to the TCA cycle and connected pathways.

## Finding 7 — Very-high-confidence AlphaFold model supports a well-ordered two-domain fold

The AlphaFold DB model **AF-Q88FB3-F1** (294 residues) is of **very high confidence**: global pLDDT of 96.56, with per-residue analysis showing mean pLDDT 96.5, **90.8% of residues at very-high confidence (>90)**, 99.3% confident (>70), and **0% low confidence (<50)**. This strongly supports a well-ordered, canonical SCS- $\alpha$  architecture in the actual *P. putida* protein: an N-terminal CoA-binding Rossmann-like domain (matching the UniProt CoA-binding domain, residues 4–100) and a C-terminal domain bearing the active-site histidine (His247). The absence of disordered regions is consistent with a stable, folded metabolic enzyme.

## Mechanistic Model / Interpretation

### The catalytic cycle

Succinyl-CoA synthetase operates through a **three-step, ping-pong-like mechanism** involving a covalent phosphohistidine intermediate on the  $\alpha$ -subunit:

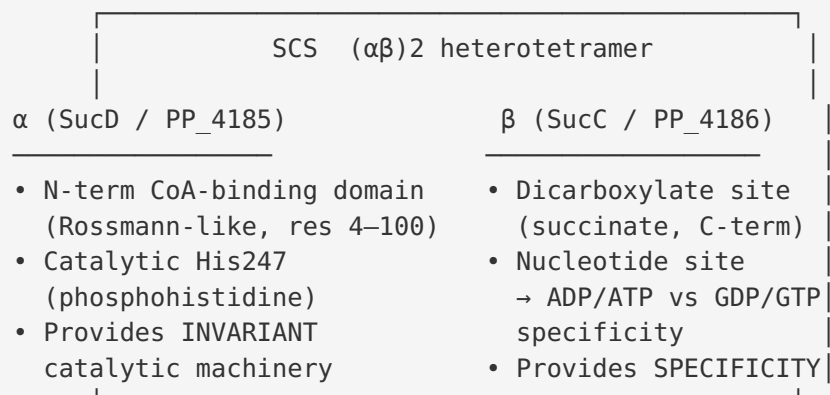
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Step 1: succinyl-CoA + Pi → succinyl-phosphate + CoA
Step 2: succinyl-phosphate → succinate + P~His (phospho-enzyme)
Step 3: P~His + NDP (ADP) → His + NTP (ATP)
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Net:    succinyl-CoA + Pi + ADP ⇌ succinate + CoA + ATP

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The **catalytic His247 of the  $\alpha$ -subunit (SucD)** is the phosphoryl carrier. Because the CoA/succinate site and the nucleotide site are ~35 Å apart, the phosphohistidine loop must **physically swing** between the two sites, shuttling the phosphoryl group. Two glutamate residues (Glu208 $\alpha$  and Glu197 $\beta$  in the studied ortholog) are crucial for the phosphorylation/dephosphorylation of this active-site histidine ([PMID: 11781092](#)).

## Division of labor between subunits



This division is the key mechanistic insight: **SucD is the "engine" (CoA + phosphohistidine chemistry) while SucC is the "selector" (which dicarboxylate, which nucleotide).** The two subunits form an intimate  $\alpha\beta$  dimer; two of which assemble into the functional  $(\alpha\beta)_2$  tetramer.

## Physiological role in *P. putida*

*P. putida* KT2440 is an aerobic soil bacterium with a strongly oxidative, TCA-centered central metabolism. In this organism the classic glycolytic pathways are notably not part of tightly-regulated "central carbon metabolism" in the textbook sense ([PMID: 24951791](#)), placing even greater importance on the TCA cycle for energy and precursor supply. SucD/SucC sit at the succinate node of this cycle, where:

- In the **oxidative TCA direction**, they conserve the thioester-bond energy of succinyl-CoA (produced by 2-oxoglutarate dehydrogenase) as ATP, regenerating free CoA and succinate (the latter feeding succinate dehydrogenase).
- Through the enzyme's **broad NTP-synthesizing capacity and coupling to Ndk** ([PMID: 10671455](#)), it can help balance the cellular pools of ATP, GTP, UTP, and CTP.
- The succinate/succinyl-CoA node connects to **propanoate** and **C5-branched dibasic acid** metabolism (KEGG ppu00640, ppu00660), and — given the  $\beta$ -subunit's "glutaryl-CoA synthetase" annotation — potentially to broader dicarboxyl-CoA metabolism relevant to *P. putida*'s renowned catabolic versatility (e.g., aromatic-compound degradation funneling carbon into the TCA cycle; [PMID: 25617773](#)).

## Evidence Base

The functional assignment rests on a **four-tier evidence hierarchy**, from most organism-specific to most mechanistically detailed:

| Tier | Evidence type                             | Source(s)                                           | What it establishes                                                                         |
|------|-------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1    | Intrinsic sequence & structure of Q88FB3  | UniProt Q88FB3; AlphaFold AF-Q88FB3-F1              | His247 phosphohistidine, N-terminal CoA domain, canonical fold in the <i>actual</i> protein |
| 2    | Genome context in <i>P. putida</i> KT2440 | KEGG ppu:PP_4185                                    | sucCD operon; TCA cycle + connected pathway assignment                                      |
| 3    | Direct biochemistry in a close relative   | <i>P. aeruginosa</i> SCS [PMID: 10671455]           | sucCD operon; broad NTP synthesis; Ndk coupling                                             |
| 4    | Mechanistic crystallography of orthologs  | <i>E. coli</i> , pig heart, <i>T. aquaticus</i> SCS | catalytic mechanism, subunit division of labor, substrate specificity                       |



## Key papers

| PMID                     | Title (abbreviated)                                                             | How it supports the findings                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">27487822</a> | <i>Structural basis for the binding of succinate to succinyl-CoA synthetase</i> | Defines SCS as the sole substrate-level phosphorylation step; localizes succinate to $\beta$ -subunit; documents strict specificity vs. citrate/fumarate/OAA |
| <a href="#">9917402</a>  | <i>A detailed structural description of E. coli succinyl-CoA synthetase</i>     | CoA binds N-terminal domain of $\alpha$ -subunit; catalytic His travels $\sim 35$ Å                                                                          |
| <a href="#">26249701</a> | <i>Structure of GTP-specific succinyl-CoA synthetase in complex with CoA</i>    | Confirms CoA + catalytic His localize to $\alpha$ -subunit                                                                                                   |
| <a href="#">9765290</a>  | <i>ATP- and GTP-specific SCS in pigeon... same alpha-subunit</i>                | Nucleotide specificity resides in $\beta$ , not $\alpha$ (SucD)                                                                                              |
| <a href="#">11781092</a> | <i>Two glutamate residues crucial for phosphorylation of active-site His</i>    | Confirms $\alpha$ -subunit catalytic histidine; identifies Glu residues for (de)phosphorylation                                                              |
| <a href="#">10671455</a> | <i>Succinyl-CoA synthetase of P. aeruginosa... broad NTP specificity</i>        | sucCD operon in Pseudomonas; broad NTP synthesis; Ndk coupling                                                                                               |
| <a href="#">22751660</a> | <i>GTP-preferring SCS from Thermus aquaticus</i>                                | $(\alpha\beta)_2$ heterotetramer quaternary structure                                                                                                        |
| <a href="#">18392745</a> | <i>Disorders caused by deficiency of succinate-CoA ligase</i>                   | Reversible reaction; ADP/GDP-forming isoform distinction; mitochondrial matrix localization (eukaryote)                                                      |
| <a href="#">24951791</a> | <i>Functional structure of central carbon metabolism in P. putida KT2440</i>    | Context for TCA-centered metabolism in the target organism                                                                                                   |
| <a href="#">25617773</a> | <i>Aromatic catabolic pathway selection... in P. putida</i>                     | Aromatic degradation funnels carbon (incl. succinate) into TCA cycle                                                                                         |

**Consistency check:** No paper contradicted the assignment. All mechanistic studies, whether on ATP- or GTP-specific isoforms, agree that the  $\alpha$ -subunit universally provides CoA binding and the catalytic phosphohistidine — exactly the components annotated in Q88FB3. The pig-heart/pigeon eukaryotic work and the bacterial *E. coli*/*T. aquaticus*/*P. aeruginosa* work are mutually consistent, differing only in localization (mitochondrial matrix vs. cytoplasm) and nucleotide preference (a  $\beta$ -subunit property).

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## Limitations and Knowledge Gaps

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- 1. No *P. putida*-specific biochemical study.** There is no direct enzymatic characterization of the KT2440 SucD/SucC enzyme itself (kinetic constants, nucleotide preference, subunit stoichiometry). The functional assignment relies on (a) the intrinsic sequence/structure of Q88FB3, (b) genome annotation, and (c) inference from the close relative *P. aeruginosa* and from more distant, structurally characterized orthologs. While these lines converge strongly, the specific catalytic parameters ( $K_m$ ,  $k_{cat}$ , nucleotide preference) of the *P. putida* enzyme remain experimentally undetermined.
  - 2. Nucleotide preference not directly measured.** The EC 6.2.1.5 [ADP-forming] annotation implies ATP formation, but the *P. aeruginosa* ortholog shows unusually broad NDP usage (ADP/GDP/UDP/CDP). Whether *P. putida* SucD/SucC is strictly ATP-forming or similarly promiscuous has not been tested directly.
  - 3. Structure is predicted, not experimental.** The AlphaFold model, while very high confidence (pLDDT 96.5), is a computational prediction. No experimental crystal or cryo-EM structure of the *P. putida* enzyme exists. Ligand-bound conformational states (open/closed) and the exact geometry of the His247 loop are inferred from orthologs.
  - 4. Physiological regulation unexplored.** How sucCD expression and SCS flux are regulated in *P. putida* under different carbon sources (e.g., during aromatic-compound catabolism) is not directly addressed. Studies in KT2440 emphasize post-transcriptional/metabolic regulation of central metabolism ([PMID: 24951791](#)), but the specific regulation of SucD is unknown.
  - 5. The  $\beta$ -subunit "glutaryl-CoA synthetase" annotation** is intriguing but unverified for moonlighting/alternative activity in this organism.
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## Proposed Follow-up Experiments / Actions

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1. **Heterologous expression and kinetic characterization.** Clone the *P. putida* KT2440 sucCD operon, purify the recombinant ( $\alpha\beta$ )<sub>2</sub> enzyme, and measure kinetic parameters (Km, kcat) for succinyl-CoA, Pi, and each NDP (ADP, GDP, UDP, CDP) to determine whether the KT2440 enzyme shares the broad NTP-synthesizing capacity documented for *P. aeruginosa*.
  2. **Site-directed mutagenesis of His247.** Mutate His247→Ala (or His247→Asn) and confirm loss of catalytic activity, directly validating the phosphohistidine mechanism in the *P. putida* protein. Complement with <sup>32</sup>P-labeling to demonstrate transient phosphorylation of the  $\alpha$ -subunit.
  3. **Experimental structure determination.** Solve the crystal or cryo-EM structure of the *P. putida* SCS, ideally with bound CoA and/or succinate, to confirm the predicted fold and the geometry of the catalytic site relative to the AlphaFold model.
  4. **Ndk interaction assay.** Test whether *P. putida* SucD/SucC physically associates with and modulates nucleoside diphosphate kinase, as observed in *P. aeruginosa*, using pull-down or co-purification approaches.
  5. **Gene deletion / conditional knockdown.** Construct a  $\Delta$ sucCD strain (or conditional mutant) and assess growth phenotypes on TCA-cycle-dependent carbon sources (e.g., succinate, aromatic compounds) and rescue by succinate supplementation, paralleling the succinyl-CoA-synthetase knockout approach used in *Toxoplasma* ([PMID: 18336823](#)).
  6. **Flux analysis under aromatic catabolism.** Given *P. putida*'s biotechnological importance for lignin valorization ([PMID: 25617773](#)), quantify SCS flux and the succinate node under different aromatic feedstocks to inform metabolic-engineering strategies.
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## Conclusion

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The gene symbol **sucD** is **unambiguous** for this target: Q88FB3 / PP\_4185 encodes the  **$\alpha$ -subunit of succinyl-CoA synthetase (succinate-CoA ligase [ADP-forming], EC 6.2.1.5)** of *Pseudomonas putida* KT2440. It catalyzes the reversible, ATP-generating conversion of **succinyl-CoA + Pi + ADP  $\rightleftharpoons$  succinate + CoA + ATP** — the TCA cycle's sole substrate-level phosphorylation. The  $\alpha$ -subunit specifically provides the **CoA-binding site** and the **catalytic phosphohistidine (His247)**, while the partner  $\beta$ -subunit (SucC/PP\_4186) supplies dicarboxylate and nucleotide specificity. The enzyme functions as a **soluble, cytoplasmic**

**( $\alpha\beta$ )<sub>2</sub> heterotetramer** encoded in the **sucCD operon**, participating in the TCA cycle and connected propanoate / C5-branched-dibasic-acid metabolism, and in pseudomonads couples to nucleoside diphosphate kinase with broad NTP-synthesizing capacity. All evidence tiers — intrinsic sequence, high-confidence structure prediction, genome context, and orthologous biochemistry/crystallography — converge on this assignment without contradiction.

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